

Our Digital World — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split; there is one line per mark. Almost every mark lost on this unit is lost to the same few confusions: trolling for catfishing, a credit line for permission, a firewall given the antivirus's job, and the belief that deleting something makes it gone. Brute force is the one place where numbers decide: the right pattern is that each extra character multiplies the work, and every figure on these papers was computed in the generator, not typed. A recurring habit worth marking down: naming a check, a guard or a licence tag with no reason. The notes under each answer name the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 (a) Rangoli photo: **A** — she chose to share it [1]
 (b) Cousin's tag: **P** — someone else posted it; Diya did nothing [1]
 (c) Shopping app recording her browsing: **P** — collected without her sharing it [1]
 (d) Sign-up form: **A** — she deliberately gave the details [1]
Row (c) is the trap: "I was using the app, so I did it." Using an app is not the same as choosing to share what it records. The letter alone earns the mark; if he writes A for (c) or (b), ask him who chose to put that data there. [4]
- 2 Posting provocative messages to upset strangers: **trolling** [1]
 Spreading a hurtful photo with a mocking caption: **dissing** [1]
 Fake identity from stolen photos: **catfishing** [1]
 Posting as a friend from a logged-in account: **fraping** [1]
The swap this question exists to catch is trolling for catfishing ("trolling is pretending to be someone"). The rows are deliberately not in the order the words are listed, so a guess by position scores badly. Fraping is about the logged-in account, not the message. Do not accept "cyberbullying" for any box. [4]
- 3 Hidden recording and sending keys: **Yes — spyware** [1]
 Tries passwords in turn: **No — brute force attack** [1]
 Encrypts files, demands payment: **Yes — ransomware** [1]
 Hides in a genuine file and copies itself: **Yes — virus** [1]
A row needs both boxes. The brute force row is the discriminator: "Yes" there means he still files every threat under malware. A correct name with the wrong Yes/No earns nothing for that row. "Virus" written for the ransomware row is the lazy default; award nothing.
- 4 (a) First failed check, **with a quote from the post** — for example Hurry: "Forward to 10 groups TODAY" [1]
 (a) Second, different failed check with its quote — for example Source: "Scientists in Germany" (no name, no university); Evidence: "PROVED" with no study named; Hurry: "or you will lose marks"; Date: the post has none [1]
 (b) Forwards measure how shareable a message is, not whether anyone checked it; his friends may all have believed it without checking [1]
Naming a check without the quote earns nothing for that check: the quote is the evidence. Other outlets also fails, but the exhibit only says the search found no study, so accept it only if he explains it that way. "Ten friends sent it" as the reason it is true is the trap from the page: it is the reverse of evidence.

- 5** (a) Title and creator: “Monsoon at Cubbon Park” by Kavya Shetty **[1]** [4]
 (a) Source and licence: photoshare.example/kavya-s, Creative Commons BY-NC (all four items, in any order) **[1]**
 (b) **Not allowed [1]**
 (b) BY-NC means non-commercial use only; selling T-shirts is commercial, and giving credit meets only the BY condition, it does not cancel NC **[1]**
The second mark in (a) needs both source and licence; three of the four items earns one mark in total. In (b) the friend’s “we’ll credit her” is the trap: crediting is manners, not permission. Award the decision mark even if the reason is vague, but never the reason mark for “because it is illegal” alone.
- 6** School network blocking a website: **firewall [1]** [3]
 Database of known malware, quarantine: **antivirus [1]**
 Finding and removing a recording app: **anti-spyware [1]**
The medium-paper reminder lists the three jobs, so this is application, not recall. If he writes antivirus for the emoji app, he has matched on “find and remove” and ignored “records what you type”; the keyword is spyware.
- 7** 5 digits: **100,000 combinations (10^5) [1]** [4]
 5 digits: **100 seconds ($100,000 \div 1,000$) [1]**
 6 digits: **1,000,000 combinations (10^6) [1]**
 6 digits: **1,000 seconds ($1,000,000 \div 1,000$), or 16 minutes 40 seconds [1]**
The wrong method this table is built to expose is scaling by length: “5 digits is 5 ÷ 4 times as long, so 12.5 seconds”, or adding 10 s a row. The right row sequence is 10 s, 100 s, 1,000 s — each row ten times the one before, because each extra digit multiplies the work by 10. Accept times written as 10^2 and 10^3 . Follow-through: if the combinations are wrong but the times are correctly worked from them, award the time marks.
- 8** Deleting removes only his own copy **[1]** [2]
 The screenshot is a separate copy that belongs to the classmate; Karan cannot reach it, and it can be saved, shared or posted again **[1]**
“Because the internet is forever” on its own earns nothing; it is a slogan, not a mechanism. The page’s trap line is “deleting removes my copy, but copies others made can remain”. Both halves are needed.
- 9** One sensible action with its reason — for example, do not like, share or reply, because that adds to the pile; or save the evidence so the school and platform can act **[1]** [2]
 A second, different action with its reason — for example report it to the platform and school, or tell a trusted adult, or support the girl being mocked privately **[1]**
“Ignore it” alone is not an answer. The bystander case is deliberate: the page says that if you are only watching, you must not add to it. An action without a reason earns half; “fight the bullies” earns nothing.

Paper B — Medium

30 MARKS

- 1** **BY:** credit the creator **[1]** [4]
NC: non-commercial use only — not to make money **[1]**
ND: no changes to the work (no cropping, editing or remixing) **[1]**
SA: share your own version under the same licence **[1]**
ND and SA are swapped more than any other pair: “no derivatives” stops changes, “share alike” allows them on condition. “Non-commercial” or “not for profit” must appear for NC; “free to use” alone is a different idea and earns nothing. “Not allowed to copy” for ND earns nothing, because copying with credit is exactly what is allowed.

- 2 Virus: **copies itself** into other files (the only one that does) [1] [3]
 Spyware: runs **hidden**, records what you do and **sends it** to someone [1]
 Ransomware: **encrypts your files** and demands payment for the key [1]
"It is harmful" or "it damages your computer" earns nothing: all three do. Corrupting files is not unique to a virus, so award the virus mark only for self-copying or spreading. Ransomware answers that say "locks" or "makes files unusable" without payment or encryption get the mark only if the demand for money is mentioned.
- 3 Known virus in an attachment: **antivirus** [1] [4]
 Stranger trying to open connections: **firewall** [1]
 Cricket1 on every account: **none of the three** [1]
 Screensaver sending browsing history: **anti-spyware** [1]
The "none of the three" row is the point: the page says no guard fixes a short password. Students who write antivirus or firewall there are reaching for a tool because a tool is on the list. The rows are in a different order from the page's matcher so the pattern cannot be memorised.
- 4 (a) **Fraping** [1] [4]
 (b) Kabir says at 4:11 that it was not him, or that he left his account logged in on the library computer at 4:12 [1]
 (c) First action with its reason — for example tell the group and the teacher that it was not him, so the false message is corrected; or log out and change his password, so nobody else can use the account [1]
 (c) Second, different action with its reason — for example save a screenshot of the 4:02 message with its time, so the school can act; or report it to the platform [1]
The wrong word to expect is catfishing: someone is pretending to be Kabir. The difference is that a catfish builds a fake identity, whereas a frape borrows a real person's logged-in account; the account, not the message, is the giveaway. "Delete the message" earns nothing: save the evidence first. Changing the password counts as an action but, on its own, does nothing about the false message.
- 5 (a) Date **PASS**, and Source and Evidence both **FAIL** [1] [4]
 (a) Hurry and Other outlets both **FAIL** [1]
 (b) These details become part of her footprint; once handed over she cannot take them back or control who passes them on [1]
 (b) A scammer can use the address and phone number to contact her family, pretend to be trusted, or find her; also they can answer security questions [1]
Only Date passes: the post carries a date, and it fits an offer that ends tonight. A check passes when there is evidence, not when the post claims to be genuine; "no company named" fails Source even though the account has a name. Each (a) mark covers the rows named on its line, so one wrong row in a group loses that group's mark. In (b) the first mark is the footprint idea (it cannot be taken back) and the second is the misuse; "because they might scam you" alone earns one.
- 6 (a) **Passive**, because her cousin posted it, not Aadhyia [1] [3]
 (b) Two things a stranger could learn: her school, her class, the time she arrives at the school gate (any two) [1]
 (b) One explained misuse — for example a stranger knows where she will be at 8:10 each day; or school and class are common answers to security questions [1]
The tag is the whole question: the phrase "I did nothing" is the passive footprint. The page also warns that school names are exactly the answers to security questions; credit that link.

- 7 (a) $26 \times 26 \times 26 = 676 \times 26 = \mathbf{17,576}$ [1] [4]
 (b) $10 \times 10 \times 10 \times 10 \times 10 = \mathbf{100,000}$ [1]
 (c) 100,000 is more than five times 17,576 ($5 \times 17,576 = 87,880$), so 5 digits are harder than 3 letters, and Zara's rule "letters always win" is wrong [1]
 (d) $100,000 \div 1,000 = \mathbf{100 \text{ seconds}}$ [1]

The tempting wrong method is to multiply instead of raising to a power: $26 \times 3 = 78$ against $10 \times 5 = 50$, which seems to prove Zara right. The true figures, 17,576 against 100,000, refute her at once. Accept "about 17,600". The lesson under it: length multiplies the work, and a bigger character set only helps if the length keeps up.

- 8 (a) **10,000** [1] [4]
 (b) **3 in 10,000** ($3/10,000$), which is **0.03%** [1]
 (c) The lock limits how many guesses he can make, so the attacker can never try every combination in turn [1]
 (d) Two-factor authentication: a second proof, such as a code sent to his phone, so a guessed password alone is not enough; or a longer, unique password, which gives many more combinations to try [1]

$3 \div 10,000$ written as 3% is the usual error (misplaced decimal point): it is off by a factor of 100, so check the scale. In (c) "the attacker gets locked out" scores only with the idea of a limit on guesses. For (d) any one of the three page defences with its reason; naming two-factor authentication without saying what it does earns nothing.

Paper C — Hard

30 MARKS

- 1 (a) Data you **deliberately** share — a post, a comment, a form you fill in [1] [4]
 (b) Data collected about you **without you deliberately sharing it** — sites visited, IP address, location, cookies, posts others make about you [1]
 (c) False or **misleading** information; the person passing it on often believes it or has not checked it [1]
 (d) The creator's legal ownership of creative work, which exists **automatically** the moment it is made [1]

Examples are not definitions: "active is when you post a photo" earns nothing without "deliberately". For (c), "fake news" or "lies" is too narrow, because misinformation includes the true-but-trimmed headline and the real-but-old photo. For (d), "a © symbol" or "you have to register it" is the common wrong belief; no symbol or registration is needed.

- 2 First named check with what he would look at — for example Source: is the author named and can I check who they are [1] [3]
 Second, different check with what he would look at — for example Date: is there one, and does it fit the event [1]
 Third, different check with what he would look at — Evidence (a link, the data or the full quote), Hurry (is it pushing me to forward), or Other outlets (do independent, reliable outlets report it) [1]

Each mark needs the check and the thing looked at. "Check if it is true" and "google it" are the question restated, not a check; "does it feel real?" is not one either. Checks may be phrased in his own words, but they must not overlap: "the date" and "whether it is old" are one check.

- 3** **Catfishing:** builds a fake identity, with stolen photos and an invented life, to deceive someone [1] [4]
Trolling: posts deliberately provocative or offensive messages to upset people or start arguments [1]
Fraping: uses someone's logged-in account without permission and posts as them [1]
Dissing: spreads hurtful information, photos or rumours about someone to put them down in public [1]
Marks are for the act, in about ten words. A definition that uses the word itself, or that says only "bullying someone online", earns nothing. The most common error is catfishing and trolling swapped; fraping answers that mention "hacking" are credited only if they say the account is already logged in or used without permission.
- 4** (a) **Allowed:** it is non-commercial and she credits the creator (BY and NC both met) [1] [5]
 (b) Allowed to change it: there is no ND tag, so cropping and adding text are permitted [1]
 (b) He must share the new poster under the **same licence** (SA), and keep the credit [1]
 (c) **Not allowed:** NC — a brochure for paid classes is commercial use, and credit does not change that [1]
 (d) **Not allowed:** BY requires credit to the creator; the fact that others do it is irrelevant [1]
Each mark needs the decision and the condition that decides it. Part (b) is the discriminating one: students who see "SA" and assume no changes are allowed have confused it with ND. Part (c) separates those who read NC as "free to download" from those who read it as "not for money". A student who answers (d) "allowed, because he found it online" has repeated the findable-is-free error.
- 5** (a) Increase = $10 - 4 = 6$; $6 \div 4 \times 100 = 150\%$ [1] [3]
 (b) Fall = $10 - 4 = 6$; $6 \div 10 \times 100 = 60\%$ [1]
 (c) It hides how small the numbers are: six extra cases. A percentage makes readers picture a wave; also accept that it hides the earlier months or what is normal [1]
The pair of parts is the point: a rise of 150% needs a fall of only 60% to undo, because the percentage is taken of a different starting number. Dividing by 4 again gives 150%, which is visibly absurd — you cannot lose more than 100% of something — so ask him how many cases a fall of 150% from 10 would leave: a negative number. Award (a) for "150" even with no working; (b) needs 10 as the divisor.
- 6** (a) 676×676 is more than $400 \times 400 = 160,000$, which is more than 100,000 (accept any valid lower bound above 316, or the exact 456,976) [1] [3]
 (b) 676 is less than 1,000, so 676×676 is less than $1,000 \times 1,000 = 1,000,000$ (or the exact 456,976) [1]
 (c) More than **100 seconds** and less than **1,000 seconds** (between 1 min 40 s and 16 min 40 s) [1]
The exact figure is $26^4 = 456,976$, and a student who works it out is correct; do not penalise. What the question rewards is bracketing by hand, between the 5-digit and 6-digit PINs. The visible wrong method is $26 \times 4 = 104$, which falls under the 100,000 line and is refuted at once by the bound in (a).

- 7 (a) **Spyware** (a keylogger) [1] [4]
 (b) **Anti-spyware**: detects and removes programs that secretly record and send your data, and can stop them installing [1]
 (c) The stolen data left as ordinary outgoing web traffic that broke none of the firewall's rules; a firewall filters traffic, it does not find or remove a program that is already installed [1]
 (d) Install software only from trusted sources; or do not install free apps from unknown sites [1]
The trap is "but I had antivirus and a firewall": two tools are on, the third job is nobody's. Do not award (b) for "antivirus", even though some products bundle it; the page keeps the three jobs separate and so must the answer. In (c), "the firewall was off" contradicts the question; award nothing.
- 8 (a) **Catfishing**: a fake account with a model's stolen photos and an invented life [1] [4]
 (a) **Dissing**: posting her secret in a group chat of 40 with a mocking caption [1]
 (b) The screenshots must show usernames, dates and times; once the chat and account are deleted the school and platform have nothing to act on [1]
 (c) Hitting back escalates it and gives the other person more to screenshot [1]
Each named act needs its part of the story. "Fraping" for the fake account is the usual wrong answer: no real account was borrowed. For (b), "save evidence" alone is thinner than "so the school and platform can act", but award the mark if the idea of keeping proof for someone to act on is clear. Deleting everything "to make it stop" is the instinct the question tests.

Paper D — Hard

30 MARKS

- 1 Cricket video: **A** — he chose to upload it [1] [4]
 Coaching app logging his location: **P** — collected without him sharing it [1]
 Friend's screenshot of a deleted comment: **P** — someone else made and shared the copy [1]
 Game review under his own name: **A** — he deliberately wrote and posted it [1]
A row earns its mark only if the reason names who chose to put the data there; "A, because it is online" has the right letter and the wrong reason. The screenshot row is the hard one: the original comment was his, but the copy in the screenshot is a friend's, so it is passive. It also quietly shows why deleting is not enough.
- 2 **Firewall**: checks network traffic against rules and blocks what breaks them — cannot clean a file already on the disk [1] [3]
Antivirus: scans files and programs against a database of known malware and quarantines or deletes what it finds — cannot stop a connection being opened [1]
Anti-spyware: detects and removes programs that secretly record and send your data — cannot filter network traffic (replace a firewall) or fix a weak password [1]
A mark needs both halves. The classic slip is giving the firewall the antivirus's job; if the "does" half reads "it protects against viruses", award nothing for that guard. "Cannot do" answers that are really about the user ("cannot stop you clicking") are acceptable for the antivirus and anti-spyware lines.

- 3** (a) **Do not retaliate** = 1 and **save the evidence** = 2 [1] [3]
 (a) **Block** = 3, **report** = 4, **tell a trusted adult** = 5 [1]
 (b) Once the person is blocked you may lose sight of their messages or profile, so screenshots showing usernames, dates and times must be taken first [1]
The order is the page's: do not retaliate, save, block, report, tell. Award the first (a) mark for the first two places both right and the second for the last three all right, so one swapped pair costs a mark. A student who puts "tell a trusted adult" first is not wrong in life; it is wrong for this paper and worth a conversation. For (b), "blocking may hide their messages" is acceptable; "so you can show someone" alone is not, without the reason the proof would be lost.
- 4** (a) 10^{12} [1] [4]
 (b) $10^{12} \div 10^3 = 10^9$ **seconds** (1,000,000,000) [1]
 (c) $10^9 \div (3 \times 10^7) = 100 \div 3 = \text{about } 33 \text{ years}$ (accept 30 to 35; a 365-day year gives 31.7) [1]
 (d) Each extra digit multiplies the work by 10; six extra digits multiply it by 10^6 , a million, not by 2 [1]
Deepak's method, doubling, gives 2,000 seconds — about 33 minutes against about 33 years — which is visibly absurd once part (c) is done. In (b), $1,000 \times 10^6 = 10^9$ is also a valid route: the 12-digit time is a million times the 6-digit time.
- 5** (a) Post 1: **real** [1] [5]
 (a) Post 2: **real but misleading** [1]
 (a) Post 3: **fake or invented** [1]
 (b) The **date** check: the photograph is from 2019, so the date does not fit the event [1]
 (c) Forwards measure how shareable a message is, not whether anyone checked it [1]
One mark per ruling. Post 2 is the one students rule "fake": the photograph is genuine and the caption is not, which is why the page calls this shape real but misleading, not invented. Post 3 is the shape most people share because it asks them to; "4,000 forwards" is the page's second trap. In (b) "Source" or "Other outlets" is acceptable only if the answer also mentions the 2019 report; the date is the cleanest answer.
- 6** (a) **Cannot** use it without the owner's permission or a licence; copyright is automatic, and being easy to find is not the same as being free [1] [3]
 (b) **Can** use it without asking: copyright has ended, so it is in the public domain (a credit is still good practice) [1]
 (c) **Can** use it unchanged, with credit to the creator (BY); cannot crop, edit or remix it (ND) [1]
Part (a) is the findable-is-free trap again, now in a film song. Part (b) has the opposite error: students who have learnt "always ask" will say ask permission anyway. In (c), giving the licence name without saying that ND forbids changes earns nothing.
- 7** (a) **Ransomware** [1] [4]
 (b) Paying does not guarantee the key arrives; the files may stay locked and the owner has lost the money [1]
 (c) Ransomware encrypts files on every drive it can reach, including a drive that is plugged in, so the backup was locked too [1]
 (d) Somewhere the malware cannot reach — for example a drive kept disconnected, or a separate offline or cloud copy [1]
"Pay it, and it will be fine" is the instinct. The page's phrase is "paying does not guarantee the key arrives". Part (c) is the real test: a backup is useless if it shares the machine with the thing that has been attacked. Accept "disconnect the drive after copying" for (d).

- 8** (i) The mistake: deleting removes only **his** copy; it is not “gone for good” **[1]** [4]
(i) Correct version: a screenshot, a forward or a saved picture is a copy others hold, so deleting does not remove copies made by others, and the photo can come back **[1]**
(ii) The mistake: pretending to be someone else is not trolling — it is **catfishing** **[1]**
(ii) Correct version: trolling is posting deliberately provocative or offensive messages to upset people or start arguments, often as a stranger **[1]**

Both statements are the two most common wrong sentences in the unit, written out as another student’s work. Marking someone else’s error is the same thinking as avoiding your own, and far easier to face. Each statement is two marks: one for finding the mistake and one for the correct version; “it is wrong” without saying why earns neither.